

# MICS Plus Tabulation

## A step-by-step guide to using the application

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## About micsPlusTableR

micsPlusTableR is an R package for preparing, tabulating, reviewing, and exploring MICS Plus survey data. It provides a browser-based application and functions that can be used directly in R. R performs the calculations; the application provides controls for selecting inputs, running analyses, and examining results.

**Purpose.** The system connects the tables you want to produce with the data and preparation steps needed to calculate them. A tabulation plan (Excel) records the table definitions, while a survey-specific R script prepares the data and variables. This makes it possible to apply a supplied plan or create and adapt a plan for your own reporting and analysis needs, then repeat the calculations when inputs or specifications change.

**Inputs.** The application uses four matching inputs for a survey:

- **Household data**, supplied as an SPSS `.sav` file.
- **Household-member data**, supplied as a separate SPSS `.sav` file.
- **A tabulation plan (Excel)**, specifying table layouts, row and column groups, conditions, statistics, and display settings.
- **A preparation folder**, containing the R script and any supporting files needed to prepare the data used by the plan.

**Features.** You can obtain supplied plans and preparation materials, prepare survey data, calculate a selected table, review consistency checks, and run several tables together. The application also lets you inspect prepared data, explore frequencies and cross-tabulations, and combine calculated table cells into a data file for further use. Worksheet filters, weights, statistics, and display settings control the tables you define.

**Results and outputs.** Choose the form that serves your task:

- **Results in the application:** table previews, check results, prepared data views, and exploratory analyses. Looking at a table in Tabulator may be all you need; writing it to Excel is not required.
- **Excel table workbooks:** save calculated tables in their worksheet layouts for review, reporting, or sharing.
- **Combined table data:** download calculated cells in Excel, CSV, or RDS format for further analysis or exchange.
- **R objects:** use the package functions in a scripted workflow.

These are different ways to use the system, not a single sequence that must end in an export. Choose the relevant tools after preparing your inputs. Save or download results when you need to retain them beyond the current app session.

## Ways to use the system

Choose the route that matches how you obtain or develop your table definitions. Either route can support viewing results in the app, reviewing them, or producing files.

### Two main routes

1. **Run an existing tabulation plan.** Use the plan and preparation files supplied by the MICS team or your survey team, together with the corresponding data. This route is for users producing tables from an agreed specification. You can run, check, and export the tables without writing calculations or changing the preparation code.
2. **Create or adapt a tabulation plan.** Design the tabulation plan for the tables you need, then apply it with a compatible preparation script and the corresponding data. This route is for plan authors and analysts who define row and column groups, statistics, filters, weights, and display settings. Use an existing working plan as a starting point where possible, keeping the worksheet structure the app expects. Check that variables used by the plan are available after preparation. If new or changed variables are needed, update the preparation script with the person responsible for that code.

**Different users, shared checks.** Table producers can use the app instructions for the tasks they need. Plan authors should also use the [worksheet reference](#) for filters and display settings. Reviewers can focus on the plan's definitions, calculated results, consistency checks, and exported tables. Users maintaining preparation scripts or running the workflow directly in R can use the package reference linked under [Terms and reference information](#). One user may take several of these roles.

After changing a plan or preparation script, load the updated inputs and rerun the affected tables and checks. Users can move between table production, plan development, and data exploration as their work requires.

## Find the task you need

The entries below cover the whole guide and link to its sections. The numbers help you find instructions; they do not mean every task must be completed. Set up the system and prepare the inputs you need, then choose the tools that serve your purpose.

### Set up the system and inputs

1. [Set up your computer](#) - install R and an editor such as RStudio.
2. [Create or open a survey project](#) - choose where your work belongs.
3. [Install the package and start the app](#) - install micsPlusTableR in R, then open the app in your web browser.
4. [Gather your survey files](#) - obtain or prepare a compatible plan, preparation script, and survey data.
5. [Load and prepare your data](#) - load your inputs and prepare the data.

### Calculate, review, explore, and save results

6. [Calculate and look at one table](#) - use Tabulator to calculate a worksheet and inspect its result.
7. [Check the table](#) - compare related counts and percentages with Consistency Checks.
8. [Save the reviewed table to Excel](#) - write a table when you need a saved workbook.
9. [Run several tables together](#) - calculate and write selected sheets, then review the run summary.
10. [Download a data file](#) - combine calculated table cells and download them as Excel, CSV, or RDS.
11. [Look at prepared data](#) - inspect selected household or household-member variables in Data View.
12. [Explore a question](#) - examine frequencies or cross-tabulations with Data Exploration.

### Manage your session

13. [Finish, return later, or get help](#) - retain any results you need, close or reopen the app, and troubleshoot.

### Reference sections

- [Worksheet reference for plan authors](#) - understand filter scope and choose decimal places.
- [Other editors and code systems](#) - use the package with an R environment other than RStudio.
- [Terms and reference information](#) - find definitions, notes about the screenshots, and links to the package reference.

# 1. Set up your computer

## Choose where to run R

If you are new to R, follow the **RStudio Desktop steps below**. R does the calculations; RStudio gives you a window in which to start them. These are two separate programs. The tabulation application then runs in your web browser. You only need to copy a few commands; the table-making steps use buttons.

An **IDE** is a program that brings together tools for working with code. The package can run in an R session started from different editors and code systems. You only need one of these options:

| Option                               | What you need to know                                                                                |
|--------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>RStudio Desktop</b>               | The route explained here, with an R Console and a project menu. Install R separately.                |
| <b>Positron</b>                      | An alternative with built-in R support. Install R, open your survey folder, and choose an R session. |
| <b>Visual Studio Code (VS Code)</b>  | Install R and configure the R extension; run the commands in an R terminal.                          |
| <b>Cursor</b>                        | Use its code editor with R installed and a compatible R extension, or start R in its terminal.       |
| <b>JupyterLab / Jupyter Notebook</b> | Use an R kernel, such as IRkernel. This requires extra setup.                                        |
| <b>R's own console / a terminal</b>  | Run the same R commands from your survey folder. No separate IDE is required.                        |

For setup details, see [Positron's R guide](#), [VS Code's R guide](#), [Cursor's extension guide](#), and [IRkernel for Jupyter](#). The [other environments](#) section explains project folders and browser access for these routes. The remaining click-by-click computer setup instructions use RStudio Desktop.

## Install R, then RStudio Desktop (once)

If both programs already open on your computer, go to [Create or open your survey project](#). You need internet access for installation. Use a recent R version supported by your editor; the package requires **R 4.2 or later**. You also need a web browser and an Excel viewer to review the saved workbooks. The workbook preview may also need internet when the app opens; see [Internet access and the workbook preview](#) for the role of browser caching.

### Install R first

1. Open your web browser and go to the [official R download site](#).
2. **Windows:** click **Download R for Windows**, then **base**, then the download link for R. Open the downloaded **.exe** file from **Downloads**. Follow the installer, using its standard options, and click **Finish**.
3. **Mac:** click **Download R for macOS**. Choose the **.pkg** installer matching your Mac and macOS version. **Apple menu > About This Mac** shows whether your Mac has an Apple chip or an Intel processor. Open the downloaded file and follow the installer.
4. **Linux:** follow the R download page's instructions for your Linux distribution.

### Install RStudio Desktop next

1. Go to [Posit's RStudio Desktop download page](#). Choose the **free RStudio Desktop** download for your operating system.
2. **Windows:** open the downloaded installer and follow its steps.  
**Mac:** open the downloaded **.dmg** and drag RStudio into **Applications**.  
**Linux:** use the installer and instructions for your distribution.
3. Open **RStudio** from the Windows **Start** menu or Mac **Applications**. If it asks you to choose R, select the version you just installed.
4. Find the **Console** panel. When you see a **>** prompt, R is ready.

**Ready to continue:** RStudio opens and its Console shows **>**. If you see "R not found," finish installing R, close RStudio, and reopen it. If your computer requires an administrator password you do not have, ask your IT support person to complete the installation.

## 2. Create or open your survey project

**Do this before starting the application.** Create or open a project folder for your survey work. It keeps your files and generated results together. Choose the appropriate route below and use a memorable name, such as **Mongolia-Wave1**; substitute your own survey.

### New work: create a new project folder

1. In RStudio, click **File > New Project**.
2. Choose **New Directory**, then **New Project**.
3. Under **Directory name**, enter your survey project name.
4. Beside **Create project as subdirectory of**, click **Browse** and choose a location you can find again, such as **Documents**.
5. Click **Create Project**. RStudio creates and opens the folder.

### You already have a folder, but no .Rproj file

1. Click **File > New Project > Existing Directory**.
2. Click **Browse** and choose your existing survey folder.
3. Click **Create Project**. RStudio adds a project file to that folder; your existing survey files stay in place.

### You already have an RStudio project

1. Click **File > Open Project**.
2. Find your survey folder, select its **.Rproj** file, and click **Open**. You can also double-click that file in File Explorer or Finder.

These routes follow the [RStudio project instructions](#). To switch surveys later, stop the app and use **File > Open Project** again.

**Check your location:** the project name should appear at the top right of RStudio.

**Your files:** keep your supplied survey data, tabulation plan, and complete preparation folder inside an **inputs** subfolder if convenient. Copying files into the project does not load them into the app; you will select them with **Browse** during [Data Preparation](#). Files may also remain in another accessible folder. Keep the project available while working; extract ZIP folders first if you downloaded the inputs as ZIP folders.

The app normally creates **micsPlusTableR-output** inside your project for Excel results. Browser downloads may go to **Downloads** instead. The **Write to Excel** screen shows the actual saved workbook paths.

### 3. Install the package and start the app

#### Install the micsPlusTableR package (once)

1. Keep your survey project open. Click the **Console** beside **>**. Use the **Console**, rather than the **Terminal** tab or a search box.
2. Install **remotes** only if it is not already available in your R library. Once installed, you do not need to install it again: skip the first line on later uses. Run the second line to install **micsPlusTableR**. Paste each needed command into the Console and press **Enter**. Wait for **>** to return before running the next command:

```
install.packages("remotes")
remotes::install_github("barisr/micsPlusTableR")
```

**remotes** is the helper that installs the application package from GitHub. The second command installs **micsPlusTableR** and its required packages, including those used by the supported survey preparation scripts. This may take several minutes. Leave RStudio open while messages scroll.

3. If R asks to create a personal library, accept it. If asked to choose a CRAN download mirror, choose **0-Cloud** if offered. If a menu lists packages with newer versions, enter the number for your choice and press **Enter**:
  - **All**: update all packages listed in that menu. Choose this when setting up a new working environment or when you want to bring the listed packages up to date and have time to check your work with the updated versions. Installation may take longer.
  - **None**: keep the versions already installed. Choose this when your current setup works and you want to keep those versions while continuing an analysis. Required packages that are missing will still be installed.

**All** refers to the packages listed for this installation, not every package available for R. If installation reports that a named package is too old, update that package, or choose **All** to update all listed packages, then retry. You do not need to update everything to resolve a problem with one package. See the [package-update guidance](#).

4. If an error appears, copy the message and ask your IT support for help. For folder or lock errors, see [Installation cannot finish](#).

You normally install once for the R library you use, not once per survey or on every launch. A different R installation or an isolated project library may need its own package installation. To update, finish and save your work and stop the app. Run the GitHub installation command again, choose package updates as described above, and restart R before using the updated package.

## Start the application (each time you work)

1. Open your survey project in either of these ways: double-click its **.Rproj** file in File Explorer or Finder, or in RStudio choose **File > Open Project**, select the **.Rproj** file, and click **Open**. Confirm the project name at the top right of RStudio. You do not need to repeat installation for ordinary launches.
2. In the **Console**, paste this line and press **Enter**:

```
micsPlusTableR::run_app()
```

3. The application opens in a browser on **Data Preparation**. On a narrow window, use **Toggle navigation** to see the tabs. If you need plans or preparation files, open **Download Files**.
4. [Gather your survey files](#), then use [Data Preparation](#) to select the country and wave, upload the files, and click **Set**.

**You are ready when:** you can open **Download Files**, **Data Preparation**, **Tabulator**, **Consistency Checks**, and **Write to Excel** across the top. Click **User's Guide** to read these instructions inside the application. Use **Open full guide** there if you want a separate browser tab.

**While working:** keep RStudio (or your chosen R environment) and the application open. A busy Console without a > prompt is normal while the app runs. Do not paste more commands there. Do not refresh the browser; you may lose work not yet written to Excel.

**No browser window?** Find the address beginning `http://127.0.0.1:` in the Console. Copy the complete address into your browser's address bar and press **Enter**. The number at the end can change each time you start.

**Where the work happens:** with this local setup, the browser connects to R on your computer. Selecting input files loads them into the running app; it does not install them into the package. Opening a project later does not restore the app's uploaded inputs or unsaved results.

If the app does not start, read the Console message and check the [package setup instructions](#). Include the complete message when requesting technical help.



## Internet access and the workbook preview

**Why the preview uses a browser library.** The workbook preview in **Data Preparation** shows the uploaded tabulation plan (Excel) inside the app. A browser cannot display an Excel workbook as a table on its own, so the app uses a JavaScript library called **SheetJS** to read it and convert the selected sheet into a displayable table. You do not need to install this library yourself. It is separate from the calculations performed by R and the calculated table preview in **Tabulator**.

**When internet is needed.** When the app opens, the browser loads SheetJS from **cdn.jsdelivr.net** if it cannot reuse a saved copy. In a local RStudio session, the workbook itself is read from the running app on your computer; the preview code does not send it to that external server. Once the library has loaded, changing the preview sheet does not require downloading the library again while that app page stays open.

**What caching means for later use.** The browser may keep a copy of the library in its cache and reuse it on later visits, avoiding another download. However, caching is **not a guarantee of offline access**: the copy can be cleared or expire, and another browser or a private window may not have it. Browser settings can also require a new request. See [how browser caching works](#).

**Before working offline:** open the app while connected, upload the plan, and check that its workbook preview appears. Keep that app page and R session open. Calculations and Excel output do not depend on SheetJS; with the required packages and input files available locally, they do not need this external download. Installation, **Download Files**, and retrieving cloud-only inputs still need internet. If only the workbook preview fails, you can inspect the plan in Excel and continue tabulating.

## 4. Gather your survey files

You need two sets of files for the **same survey (country, period, and wave)**:

- **Household and household-member data (.sav):** obtain them from the [MICS surveys page](#) with the required permission. These data are never distributed through this files repository or the package's download functions.
- **Tabulation plans and preparation files:** use a compatible set you have created or adapted, or download supplied materials from [micsPlusTableR-Files](#) using the app or one of the alternatives below. These downloads require no GitHub account once the repository is public.

If your inputs are already available, go to [Check your input set](#). Otherwise, use one of the download routes below for supplied plans and preparation files.

### Download with the app

Surveys are named by country, survey period, and wave, for example **Jamaica (2023-24) Wave 1**. The download choices and survey folders use the same names. Each survey contains its tabulation plan and a **prep-files-** folder with the matching survey name. Keep the preparation folder and its contents together.

1. Open **Download Files** and click **Load / refresh surveys**. Internet access and a public files repository are needed to load the list.
2. To download **one** folder, keep **Selected surveys** and tick that survey, such as **Jamaica (2023-24) Wave 1**. To download **several**, tick each survey you want. For **every** published survey folder, select **All surveys**.
3. Review **Selected survey folders** for the file count and total size.
4. Check **Download folder**. It starts at the current project you opened before launching the app. For example, selecting **Jamaica (2023-24) Wave 1** saves its materials in a **Jamaica (2023-24) Wave 1** subfolder directly inside that project.
5. To choose another destination, click **Browse folders...**, select a folder, and click **Select**, or type a path. You may create a folder in the picker. A relative path such as **inputs** means a subfolder of the current project. **Use current project** restores the original default.
6. Leave **Replace existing files** off to protect local copies. If the selected files already exist, use a different folder or turn it on deliberately.
7. Click **Download files** and wait for the success message, which shows where the files were saved. The preparation subfolders are preserved.

Files are saved directly by R to the chosen location. They do not go through your browser's **Downloads** folder. In a remotely hosted R session, the path is on that remote machine; use your host's file access tools to retrieve files.

## Download Files screen

User's Guide **Download Files** Data Preparation Tabulator Consistency Checks Write to Excel Multi-Sheet Tabulator Data Exploration Data View Long Format Data MICS Plus Tabulation

Load / refresh surveys

Which survey folders?

☒ Selected surveys

☐ All surveys

Choose one or more surveys

Download folder

/Users/barisucar/Documents/projects-4/mic

Browse folders... Use current project

Each survey gets its own subfolder in this location.

☐ Replace existing files

Download files

Download tabulation plans and preparation files

Load the available surveys, choose one or more folders (or all surveys), then choose where to save them.

Household and household-member data are obtained separately from the [MICS Plus website](#) with the required permission. They are not included in these downloads.

Current project: /Users/barisucar/Documents/projects-4/micsPlusTableR-manager

Selected survey folders

No survey list loaded yet.

Click Load / refresh surveys to see the published materials.

After downloading

Open Data Preparation and select the downloaded Excel plan and complete preparation folder, together with your separately obtained household and household-member data.

[Alternative: download all plans and preparation files as a ZIP](#)

This picture shows the screen before **Load / refresh surveys** is clicked. Your own project path appears in **Download folder**. After loading, the available survey checkboxes appear under **Choose one or more surveys**.

## Alternative: download a ZIP through your browser

1. Click [Download all plans and preparation files \(ZIP\)](#).
2. Find the ZIP in **Downloads**. On Windows, right-click it and choose **Extract All**; on a Mac, double-click it.
3. Open the extracted `micsPlusTableR-Files-main` folder. Copy the survey folder you need, such as `Jamaica (2023-24) Wave 1`, into your survey project's **inputs** folder. Keep its preparation folder and all subfolders together.

## Alternative: download from the R Console

Open your survey project. Run these commands in the **R Console before starting the app**. If the app is running, stop it first so the `>` prompt returns. First list the available surveys:

```
available <- micsPlusTableR::list_survey_files()
unique(available$survey)
```

Choose **one** of these alternatives:

```
# One survey
micsPlusTableR::download_survey_files("Jamaica (2023-24) Wave 1")

# Several surveys
micsPlusTableR::download_survey_files(c(
  "Jamaica (2023-24) Wave 1",
  "Mongolia (2025-26) Wave 2"
))

# All published surveys
micsPlusTableR::download_survey_files()
```

The Console function defaults to **inputs** under `getwd()`, while the app tab defaults to the project itself. Console downloads preserve the survey and preparation subfolders. Use `dest_dir = "other-folder"` to choose another destination. Existing files are protected: use a new folder, or add `overwrite = TRUE` only when you intend to replace them.

For individual files, inspect `available$path` and pass the exact paths:

```
micsPlusTableR::download_survey_files(
  files = available$path[1],
  dest_dir = "selected-inputs"
)
```

Select a whole survey folder to include its preparation helpers. Add the separately authorized household/member data before running the app. If GitHub limits requests, retry later or use the ZIP link.

## Check your input set

Keep these **four items for the same country, period, and wave** together in your survey project's **inputs** folder, or another folder you can find again. The household and member files come from the MICS Plus website with permission. Use a tabulation plan and preparation folder supplied by your team or downloaded from [micsPlusTableR-Files](#), or your own compatible versions. Once all four items are ready, continue to [Data Preparation](#) in the app.

| Item to obtain         | How to recognize it                                               | Where it goes in the app             |
|------------------------|-------------------------------------------------------------------|--------------------------------------|
| Household data         | Name often ends in <b>hh.sav</b>                                  | <b>Select household file</b>         |
| Household members data | Name often ends in <b>hl.sav</b>                                  | <b>Select household members file</b> |
| Tabulation plan        | Excel file ending in <b>.xlsx</b>                                 | <b>Select tabulation file</b>        |
| Preparation folder     | Folder containing the preparation script and its supporting files | <b>Select preparation folder</b>     |

Your team may supply longer names. Do not rename the files just to match the pictures. When running an existing plan, you do not need to open the **.sav** files or edit the preparation code. When creating or adapting a plan, check its variable names and definitions against the prepared data and script before running it.

## 5. Load and prepare your data

Open **Data Preparation** in the app. Use the controls below to select your files and preparation folder, then follow **Run Data Preparation** to check the survey choices and click **Set**.

### How to choose a file

1. Click **Browse...** beside the relevant item in the application.
2. In the window that opens, find the folder containing your inputs. If it was downloaded, first look in **Downloads**.
3. Double-click a folder to open it. Click the required file once to select it.
4. Click **Open**. Wait until its name appears in the application and the upload finishes. Repeat for the other files.

### How to choose the preparation folder

1. If the team supplied a ZIP file, extract it first: on Windows, right-click it and choose **Extract All**; on a Mac, double-click it.
2. Beside **Select preparation folder**, click **Browse...**
3. Select the folder that contains the preparation file, then click **Upload**, **Select**, or **Open**, depending on your browser. Select the folder itself, rather than an individual file inside it.
4. If the browser asks to upload the folder's files, check its name and confirm.

**Check before continuing:** use the complete preparation folder for the selected survey. Keep any folders inside it, such as **FIES-inputs**, in place. Check the preparation script's file references if you are unsure which supporting files belong with it.

## Run Data Preparation

The screenshot displays the 'Data Preparation' tab in the MICS Plus Tabulation application. The left sidebar contains several sections for file selection:

- Select country and period:** A dropdown menu showing 'Mongolia (2025-26)'.
- Select wave:** A dropdown menu showing 'Wave 2'.
- Select household file:** A 'Browse...' button next to 'MNG\_W2\_hh.sav' with an 'Upload complete' status.
- Select household members file:** A 'Browse...' button next to 'MNG\_W2\_hl.sav' with an 'Upload complete' status.
- Select tabulation file:** A 'Browse...' button next to 'MNG MICSPPlus 2025-26 V' with an 'Upload complete' status.
- Select preparation folder:** A 'Browse...' button next to 'prep-files-MNG\_W2' with an 'Upload complete' status.

Below these sections, the 'Main prep' information is shown: 'MNG MICSPPlus 2025-26' and 'Wave2\_prep-data.R | FIES helpers: none'. A 'Set' button is located at the bottom of this section.

The right panel is divided into two main sections:

- Survey info consistency:** A message with a green checkmark stating 'The information from the Reference file is consistent.' Below this, the reference information is listed: 'Reference (IDX): Country: Mongolia (2025-26) Wave : 2'.
- Tabulation Plan:** A table showing the mapping of survey tables to healthcare service tables. The 'Sheet' dropdown is set to 'IDX'.

| Mongolia                                               |                                                                                                              |
|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| MICS Plus                                              |                                                                                                              |
| 2025-26                                                |                                                                                                              |
| Wave 2                                                 |                                                                                                              |
| Content <sup>A</sup>                                   |                                                                                                              |
| Survey and sample characteristics                      | Healthcare services: utilization & expenditures                                                              |
| Table 1.1 Response rates (unweighted)                  | Table 6.1 Healthcare services: Need and access                                                               |
| Table 1.2 Average number of call attempts (unweighted) | Table 6.1a Healthcare services: Access to healthcare for children                                            |
| Table 1.3 Number of call attempts (unweighted)         | Table 6.1b Healthcare services: Types of healthcare services not received and main reasons for not receiving |
| Table 1.4 Duration of interviews (unweighted)          | Table 6.2 Healthcare services: Types of healthcare services received                                         |
| Table 2.1 Housing characteristics                      | Table 6.2a Healthcare services: Expenses for healthcare services received                                    |
| Table 2.1a Household and personal assets               | Table 6.3 Healthcare services: Access to and use of medicines                                                |
| Table 2.1b Wealth quintiles                            | Table 6.4 Healthcare services: Hospitalization and related expenses                                          |
| Table 2.2 Household composition                        |                                                                                                              |

At the bottom of the left sidebar, an 'Excel output folder' section indicates that generated output and formatted workbooks are written to: `/Users/barisucar/Documents/projects-4/micsPlusTableR-manager/micsPlusTableR-output`.

Figure 1: Data Preparation: Mongolia Wave 2 inputs are selected on the left; the survey consistency result and IDX sheet of the tabulation plan appear on the right.

1. Click **Data Preparation**. Under **Select country and period**, click the small arrow and choose your survey. Choose the matching **wave** below it.
2. Use the four **Browse...** controls to [choose the files](#) and [select the preparation folder](#).
3. Read **Main prep** below the folder control. Check that it names the intended preparation file. **FIES helpers: none** is normal for surveys without FIES.
4. Click **Set** once. If it is below the visible area, scroll down inside the left panel. Wait for preparation to finish.
5. Read **Survey info consistency** at the upper right. Check the country, period, and wave. The example says the reference information is consistent.

**Ready to continue:** the survey details agree and there is no preparation error. A matching survey message alone does not resolve a separate error. If a mismatch or error appears, compare the selected country and wave with the tabulation plan's IDX sheet, and inspect the reported preparation error before continuing.

## 6. Calculate and look at one table

User's Guide Download Files Data Preparation **Tabulator** Consistency Checks Write to Excel Multi-Sheet Tabulator Data Exploration Data View Long Format Data

MICS Plus Tabulation

Select a sheet

1.1

Format type

☐ Unformatted
 ☐ Suppressed
 ☒ Formatted

Pivot type

☐ Index
 ☒ Header
 ☐ Logic

Run

Table name

Table 1.1 Response rates (unweighted)

Pivot table preview

|                                                                                      | Area  |       |       | Region  |         |         |         |             | Degree of urbanization |                        |            |
|--------------------------------------------------------------------------------------|-------|-------|-------|---------|---------|---------|---------|-------------|------------------------|------------------------|------------|
|                                                                                      | Total | Urban | Rural | Western | Khangai | Central | Eastern | Ulaanbaatar | City                   | Town & Semi-dense area | Rural area |
| (Households sampled) Households sampled                                              | 2,903 | 1,501 | 1,402 | 536     | 528     | 579     | 574     | 686         | 586                    | 902                    | 1,415      |
| (Households sampled) Interviewed                                                     | 2,264 | 1,172 | 1,092 | 429     | 422     | 457     | 417     | 539         | 451                    | 705                    | 1,108      |
| (Households sampled) No consent for next wave                                        | 188   | 114   | 74    | 27      | 34      | 43      | 32      | 52          | 40                     | 76                     | 72         |
| (Households sampled) With consent for next wave                                      | 2,076 | 1,058 | 1,018 | 402     | 388     | 414     | 385     | 487         | 411                    | 629                    | 1,036      |
| (Households sampled) Refused                                                         | 178   | 112   | 66    | 22      | 29      | 36      | 43      | 48          | 43                     | 62                     | 73         |
| (Households sampled) No consent for next wave                                        | 169   | 106   | 63    | 22      | 29      | 31      | 42      | 45          | 40                     | 58                     | 71         |
| (Households sampled) With consent for next wave                                      | 9     | 6     | 3     | 0       | 0       | 5       | 1       | 3           | 3                      | 4                      | 2          |
| (Households sampled) No eligible respondent                                          | 0     | 0     | 0     | 0       | 0       | 0       | 0       | 0           | 0                      | 0                      | 0          |
| (Households sampled) Phone number(s) does not belong to sampled household            | 33    | 12    | 21    | 6       | 6       | 7       | 10      | 4           | 2                      | 12                     | 19         |
| (Households sampled) Phone number(s) inactive                                        | 42    | 15    | 27    | 7       | 4       | 12      | 13      | 6           | 4                      | 12                     | 26         |
| (Households sampled) Respondent busy / postponed                                     | 132   | 62    | 70    | 21      | 26      | 26      | 32      | 27          | 27                     | 39                     | 66         |
| (Households sampled) No response after repeated call attempts or phone(s) turned off | 157   | 60    | 97    | 41      | 27      | 23      | 42      | 24          | 22                     | 37                     | 98         |
| (Households sampled) Other                                                           | 0     | 0     | 0     | 0       | 0       | 0       | 0       | 0           | 0                      | 0                      | 0          |
| (Households sampled) Not called in the current wave                                  | 97    | 68    | 29    | 10      | 14      | 18      | 17      | 38          | 37                     | 35                     | 25         |
| (Response rate) Response rate                                                        | 78.0  | 78.1  | 77.9  | 80.0    | 79.9    | 78.9    | 72.6    | 78.6        | 77.0                   | 78.2                   | 78.3       |

1. Click **Tabulator** at the top.
2. Under **Select a sheet**, click the arrow. Choose the table code that you require. The picture uses **1.1**.
3. Leave **Format type** on **Formatted** and **Pivot type** on **Header** for an easy-to-read preview.
4. Click **Run** once and wait for the table to appear.
5. Read **Table name** above the result. Check that it is the table you intended to produce. Scroll inside the preview to see more rows or columns.

**Ready to continue:** the correct table name and calculated values appear. Use **Consistency Checks** to review the applicable comparisons. If viewing the result meets your need, you do not have to export it. Use **Write to Excel** when you want to keep a table workbook; the preview itself does not save one.

If you require plain numbers, choose **Unformatted**. Choose **Suppressed** to inspect the small-sample display rules.

**Choose the labels with Pivot type:**

| Option        | What it shows and when to use it                                                                                                              |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Header</b> | Descriptive row and column labels from the tabulation plan. Use this to read and interpret the table.                                         |
| <b>Index</b>  | The worksheet's row and column numbers. Use this to locate a result in the tabulation plan; for example, row 9 and column 6 identify cell F9. |
| <b>Logic</b>  | The row and column conditions used for the table. Use this to inspect the expressions behind the groups being compared.                       |

Pivot type changes the preview labels, not the calculated values. Choose the view you need, then click **Run** to refresh the preview.



## 7. Check the table

Table name

Table 1.1 Response rates (unweighted)

Row Group (N) Row Group (Perc) Row Indent Group (N) Row Indent Group (Perc) Col Group (N)

Col Group (Perc)

Search

| var_name_row | cond | col_index | stat_type | group_total | total_value | diff_value |
|--------------|------|-----------|-----------|-------------|-------------|------------|
| HH17         | 1    | 3         | n_unw     | 2903        | 2903        | 0          |
| HH17         | 1    | 4         | n_unw     | 1501        | 1501        | 0          |
| HH17         | 1    | 5         | n_unw     | 1402        | 1402        | 0          |
| HH17         | 1    | 7         | n_unw     | 536         | 536         | 0          |
| HH17         | 1    | 8         | n_unw     | 528         | 528         | 0          |
| HH17         | 1    | 9         | n_unw     | 579         | 579         | 0          |
| HH17         | 1    | 10        | n_unw     | 574         | 574         | 0          |
| HH17         | 1    | 11        | n_unw     | 686         | 686         | 0          |
| HH17         | 1    | 13        | n_unw     | 586         | 586         | 0          |
| HH17         | 1    | 14        | n_unw     | 902         | 902         | 0          |
| HH17         | 1    | 15        | n_unw     | 1415        | 1415        | 0          |

1. Click **Consistency Checks**. Confirm the **Table name** matches the table you just ran.
2. Click **Run Checks** and wait for the results.
3. Click each of the smaller check tabs, such as **Row Group (N)**. They compare counts or percentages in different parts of the table.
4. If a tab is red, tick **Show only inconsistent cells** to find the differences. Inspect the affected groups, their filters, and their denominators to identify the cause.

**Green:** this check ran and found no difference beyond its tolerance. **Red:** a difference needs review. **No color or no rows:** the check may not apply; it does not mean it passed. Read the result as well as the color. The **Difference** column shows zero where the compared values agree. **Sum of group values** is compared with the **Reference total**; in column percentage checks, **Sum of percentages** is compared with 100. **Worksheet row** and **Worksheet column** identify positions in the tabulation plan. The screenshot shows the older column labels; `diff_value` is now displayed as **Difference**.

**Ready to continue:** the applicable checks pass, or you have explained and documented any expected differences. Also check that the table's population, labels, statistics, and interpretation match its purpose; matching totals alone do not establish that.

## 8. Save the reviewed table to Excel

Use **Write to Excel** when you want to keep or share tables in workbook form. You can inspect tables in **Tabulator** without creating these files.

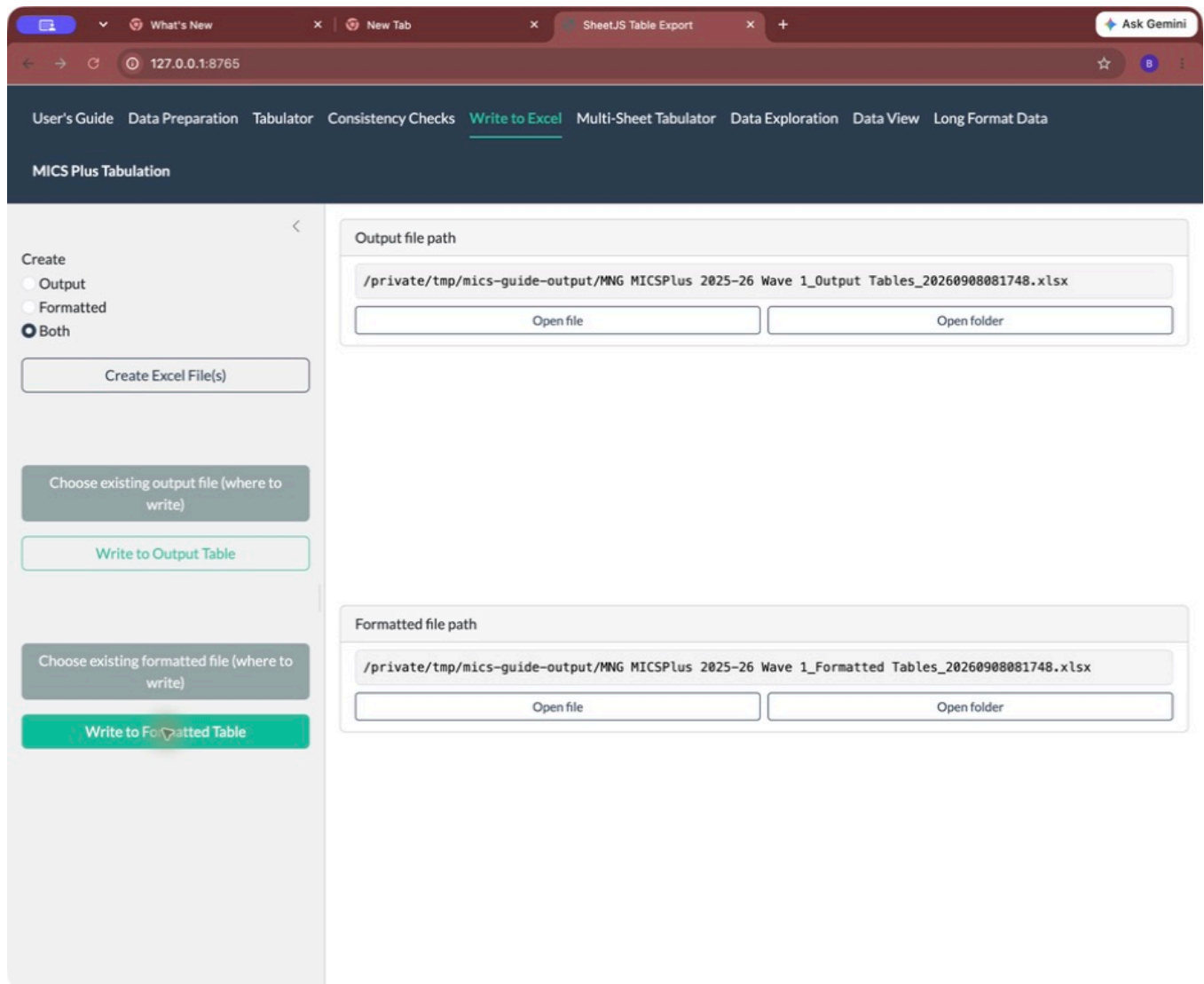


Figure 2: Write to Excel: create Both workbooks, write the current table with the two write buttons, and find the saved files using Open folder.

1. Click **Write to Excel**. Under **Create**, choose **Both** to create the Output and Formatted workbooks, or select the single workbook type you need.
2. Click **Create Excel File(s)** once for this run. Two file paths appear on the right. This creates workbook copies; it does not fill every table.
3. Click **Write to Output Table**. Wait for the write to finish.
4. Click **Write to Formatted Table**. Wait again. Both buttons write the **current table** to its matching sheet.
5. Under **Formatted file path**, click **Open folder** to find the saved file, or **Open file** to view it. Check the expected sheet in Excel.

**Output** keeps numeric values for review. **Formatted** contains the table's presentation values and applicable notes. Close the workbook in Excel before writing another table. If **Open folder** does not work, copy the displayed folder path into File Explorer or Finder.

**Next table:** repeat **Tabulator > Run, Consistency Checks > Run Checks**, then the two **Write** buttons. Reuse the same workbooks; do not click **Create Excel File(s)** again for every table.

## 9. Run several tables together

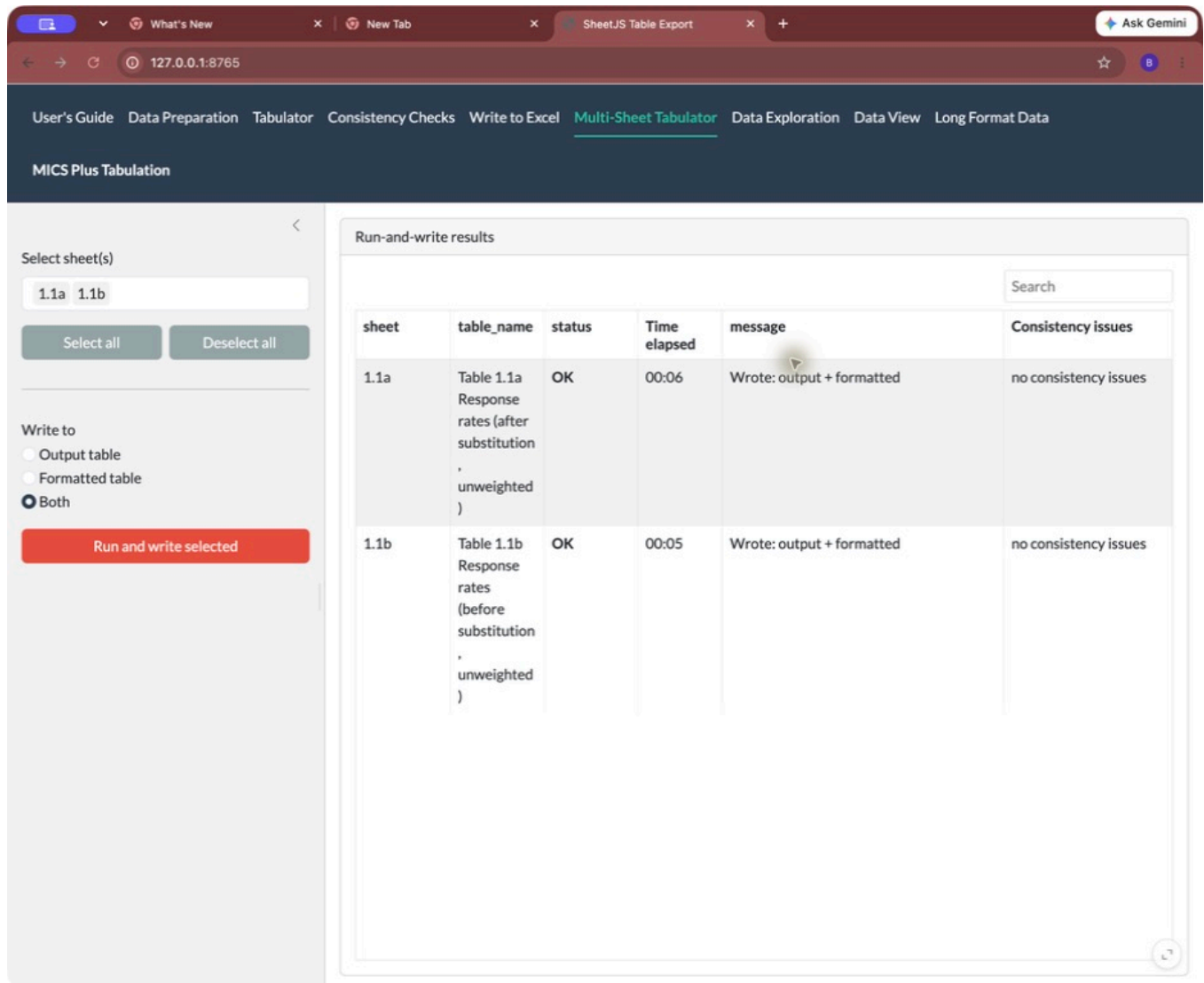


Figure 3: Multi-Sheet Tabulator: choose sheets, choose Both, then run and inspect one result row per sheet.

Prepare the data, then use **Multi-Sheet Tabulator** to calculate and write selected tables together.

1. First [create your workbook files](#) in **Write to Excel**, if you have not already done so for this run.
2. Click **Multi-Sheet Tabulator**. Click **Select sheet(s)** and choose a table. Click again to add another. Use **Deselect all** to clear the selection.
3. Under **Write to**, choose **Both** or your agreed destination.
4. Click **Run and write selected** once. Keep your R environment and the browser open until it finishes.
5. When the completion message appears, click **OK**. Read every row in **Run-and-write results**, especially **status**, **message**, and **Consistency issues**. Use the horizontal scrollbar if a column is off the right edge.

**Ready to continue:** every requested table has completed and its results have been reviewed. This action writes tables as it runs; it does not wait for your review of each table. A completed write is not approval of a flagged result.

If a table fails or has consistency issues, note its code and message. Investigate the error or difference before using the result. Start with a few tables; use **Select all** when ready for the full run.

## 10. Download a data file

The screenshot displays the MICS Plus Tabulation web application interface. On the left, the 'Long Format Data' section is active, featuring a 'Run all tables and prepare...' button and a 'Download output' section with buttons for Excel (.xlsx), CSV (.csv), and RDS (.rds). The right panel shows a preview of the 'Output Table' with columns: STRUCTURE, STRUCTURE\_ID, ACTION, FREQ, REF\_AREA, SURVEY\_ID, and TABL. The table contains 18162 rows, with the first 100 rows visible. The bottom status bar shows '127.0.0.1:8765/#tab-4346-9'.

Figure 4: Long Format Data: prepare the data with the button on the left, then use a download button below it.

Use **Long Format Data** when you need calculated table cells together in one dataset for analysis, exchange, or further processing. This produces a combined data layout; **Write to Excel** saves tables in their worksheet layouts. Choose the output that serves your purpose.

1. Complete **Data Preparation** first.
2. Click **Long Format Data**.
3. Click **Run all tables and prepare....** Wait while the application calculates the tables. This may take longer than running one table.
4. When the data preview appears without an error, find **Download output** in the left panel. Click **Excel (.xlsx)** for a spreadsheet, or the format your recipient requested.
5. Find the file in your browser's **Downloads** list or your computer's **Downloads** folder. Some browsers ask you to choose a save location.

**What you receive:** one combined file of calculated table cells. It is a different layout from the **Formatted** workbook. **CSV (.csv)** is a plain text table; **RDS (.rds)** is for someone working in R.

The download contains all prepared rows, including rows hidden by preview filters or pages. Check the country, period, and wave in the filename before sending it. If a table fails, inspect its error message, correct the relevant input or instruction, and rerun before relying on the download.

## 11. Look at prepared data

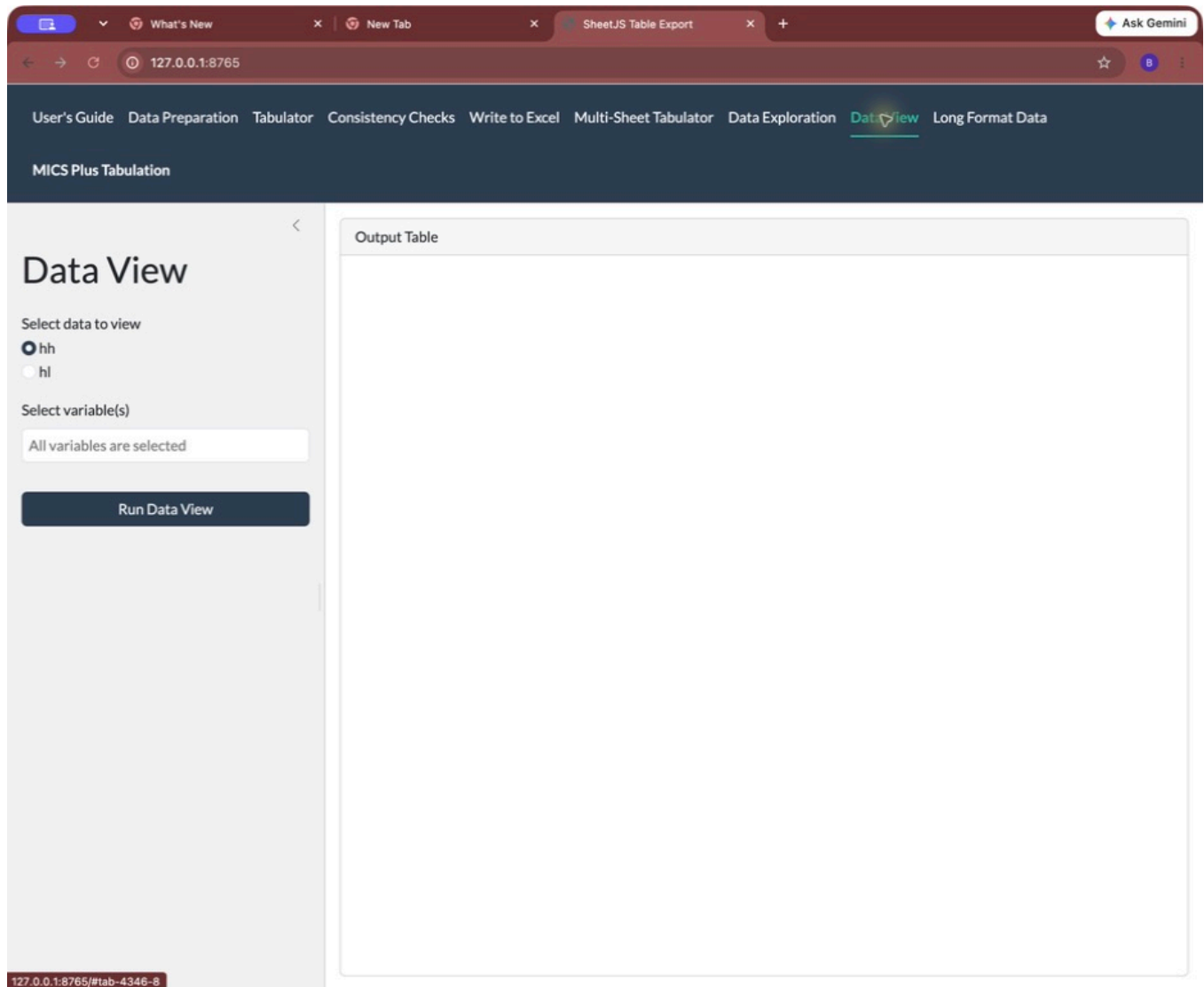


Figure 5: Data View: select the household or household members data, select variables, then click Run Data View.

**Data View** shows data after preparation. Use it to inspect variable values and labels, understand the prepared dataset, or investigate a table result. A **variable** is a named item in the data, like age or region.

1. Complete **Data Preparation**, then click **Data View**.
2. Under **Select data to view**, choose **hh** for households or **hl** for household members.
3. Select the variables you want to inspect.
4. Click **Run Data View**. Read the preview and its labels.

**Check:** you are looking at the intended dataset and variables. This screen may show individual survey records. Share only what your survey team's data handling rules allow, and avoid including records in a help screenshot.

You can return to **Tabulator** by clicking its tab at the top. Viewing data here does not save a presentation table to Excel. The picture shows the controls before any individual records are displayed.

## 12. Explore a question

Figure 6: Data Exploration: choose the dataset, variables, and analysis type before clicking Run Data Explorer.

**Data Exploration** lets you investigate questions through frequencies and cross-tabulations without changing the tabulation plan. Use it to understand the data, examine relationships, or investigate a result. These analyses are separate from the tables defined in the plan.

1. After **Data Preparation**, click **Data Exploration**.
2. Under **Select data set**, choose **hh** for households or **hl** for household members. Under **Select variable(s)**, choose the variables relevant to your question.
3. Choose **Unweighted** or **Weighted** for the analysis you intend. If weighted, enter the appropriate survey weight in **Type weight variable**. Check the preparation script or survey documentation if you are unsure which weight to use.
4. Choose **Frequency** to count categories of one variable, or **Cross table** to compare two variables. Select the second variable if needed.
5. Leave the filter box blank for all prepared records, or enter the condition for the population you want to examine. Click **Run Data Explorer** and read the result.

**Check:** the selected data, variables, and weighting match the question you are exploring. These choices affect the meaning of a result. Apply your survey team's review process before reporting a result as a final survey finding.

## 13. Finish, return later, or get help

### Finish your work

1. If you need to retain results, [write tables to Excel](#) or [download combined table data](#). If viewing results was sufficient, no export is needed.
2. If you saved files, open them and check the country, wave, and expected contents. Keep any review notes with them.
3. Close any open workbooks, then close the application's browser tab.
4. In RStudio, click the red **Stop** button in the Console to stop the app. In another editor, use its R session interrupt/stop control. You may then close the editor. Saving an R workspace is not a substitute for saving the Excel workbooks.

**When you return:** reopen the same survey project, run `micsPlusTableR::run_app()`, and select your inputs again. The app does not automatically restore the previous browser session. For a fresh run, create new workbook files if you intend to export tables. To continue an existing workbook, select it using the **Choose existing...** buttons in **Write to Excel**. Writing the same sheet again replaces its previously written values.

## If something goes wrong

| What you see                                                         | What to do next                                                                                                                                                                                                                                                               |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Installation reports <code>setwd(startdir)</code> or a 00LOCK folder | Follow <a href="#">Installation cannot finish</a> .                                                                                                                                                                                                                           |
| Download list unavailable                                            | Check internet access and that the files repository is public, then click <b>Load / refresh surveys</b> .                                                                                                                                                                     |
| Download says files already exist                                    | Choose another download folder, or deliberately enable <b>Replace existing files</b> .                                                                                                                                                                                        |
| Missing file or wrong filename                                       | Return to <b>Data Preparation</b> , select the correct file, and click <b>Set</b> again. Rerun and recheck the table.                                                                                                                                                         |
| Country or wave mismatch                                             | Compare the dropdown choices with the IDX sheet and the country, period, and wave of your input files. Select a matching set.                                                                                                                                                 |
| Preparation error or missing variable                                | Check the named variable, selected data source, and preparation step. If the script needs a change you cannot make, provide its maintainer with the complete error message.                                                                                                   |
| No calculated table in Tabulator                                     | Check that <b>Set</b> completed, choose a sheet, and click <b>Run</b> .                                                                                                                                                                                                       |
| Workbook preview alone is blank or says Preview failed               | Check your connection and whether <b>cdn.jsdelivr.net</b> is blocked. A cached library is not guaranteed; see <a href="#">workbook preview and caching</a> . You can inspect the plan in Excel and continue tabulating. Save your results before restarting the app to retry. |
| Red consistency tab                                                  | Inspect the displayed differences and the related groups, filters, and denominators. Correct errors or document expected differences before using the result.                                                                                                                 |
| Excel file cannot be written                                         | Close that workbook in Excel, then try its <b>Write</b> button again.                                                                                                                                                                                                         |
| Cannot find the saved workbook                                       | Use <b>Write to Excel &gt; Open folder</b> or the path shown there. For a data download, check <b>Downloads</b> instead.                                                                                                                                                      |
| Browser disconnected or refreshed                                    | Save any work still available. Restart the app if needed and select the inputs again; check existing files before rewriting.                                                                                                                                                  |

**When asking for help, include:** country and wave, table code, the button you clicked, and the exact error text. A screenshot of the message helps; leave individual survey records out of it. Your support person can check the installed package version in RStudio.



## Read a calculation error

Calculation errors now identify the worksheet and, where available, the filter block, Excel row and column, statistic, and condition that failed. For example:

```
Worksheet 'Example', horizontal block 2, starting column 4 ...  
Excel row 9, column 4; column condition: age >= 18; statistic: p  
object 'age' not found
```

Here, **age** is missing from the prepared data used for that block. The row and column refer to positions in the tabulation plan. A block is a group of columns that share a filter. An index alone (such as “In index: 1”) does not explain the cause; keep the complete message, including the expression and the final cause.

- **Prepared data source unavailable:** confirm that data preparation finished and that the plan names the correct household or household-member file.
- **Weight column not found or not numeric:** check the plan’s “weight by” entry against the prepared data. Confirm that the named variable exists and contains numeric weights.
- **Unknown variable or invalid expression:** use the reported cell position to check the condition’s variable names and R syntax against the prepared data.
- **Unsupported statistic:** the statistic cell contains a code that this direction of tabulation does not support. Use a statistic supported by that direction and appropriate to the intended calculation.
- **No data-source condition in B4:B7:** check that the selected worksheet is a table plan and contains its source/filter instructions in the expected cells.
- **Extra-table dimensions differ:** the preparation script’s supplied table must match the plan’s row count, with one label column and the value columns.
- **Comparison row or column counts differ:** align the two tables and check skipped header rows before comparing again.

After correcting the input, repeat data preparation if the script changed, then calculate and check the table again.

## Installation cannot finish

**Recognize the messages:** Error in `setwd(startdir)` : cannot change working directory, or failed to lock directory followed by a folder such as `00LOCK-micsPlusTableR`. Follow the recovery steps below.

**What happened:** the first message means R's installer could not return to its starting folder. If that also prevents recovery, an unfinished installation can leave a lock folder that blocks later attempts. A lock error can also mean another installation is still running. These messages alone do not show why the starting folder became unavailable or prove a fault in the application.

## Recover the installation

1. Save your work and the complete Console error. Close the app and other R sessions after any active installations finish.
2. Open your survey's **.Rproj** file from its normal, existing project folder. Keep that folder available throughout installation. Do not install from inside R's installed `micsPlusTableR` folder.
3. **Only if a lock error remains:** check the library path printed in the error and confirm no installer is running. Move only the reported stale lock folder to a separate backup location outside the library. It may contain the previous package needed for recovery. Do not delete the library or remove locks belonging to active installations.
4. Retry the original command in the Console:

```
remotes::install_github("barisr/micsPlusTableR")
```

5. After installation reports `DONE (micsPlusTableR)`, verify that it loads:

```
library(micsPlusTableR)
packageVersion("micsPlusTableR")
```

No manual `setwd()` command or special installation options are required for this recovery route. If it fails again, keep the new error and stop repeating the installation until support has investigated it.

## Prevent repeated working-directory failures

**For support staff:** run installation in a separate R process with an explicit temporary working directory outside the package library. Keep that directory alive until installation and cleanup finish, and pass the intended library paths to the process. This avoids relying on the user's current folder without changing their session's working directory. It requires a separate installation launcher; the standard GitHub command does not acquire this behavior from a change to `micsPlusTableR` alone.

Keep R's staged installation, loading checks, and locking enabled. Clearing a stale lock repairs a failed attempt; it does not fix the original cause. Maintainers should require successful installation and reinstallation checks before release and report a reproducible installer failure to R or `remotes`. This addresses the known failure mode, not every possible installation error.

For investigation, record `getwd()`, `.libPaths()`, and `sessionInfo()` before retrying. These commands only report information. Review paths for personal details before sharing the report. See the [R installation documentation](#).

## Worksheet reference for plan authors

Use this reference when creating, adapting, or reviewing a tabulation plan. It explains worksheet filters and decimal settings. Users running an agreed plan can use the app's tools without changing these instructions. After editing a plan, save and upload the updated workbook, then rerun the affected tables and checks.

### Understand worksheet filters

This section is for people who prepare or review the tabulation plan. It explains which records a `filter(...)` instruction includes when the table is calculated. These are worksheet instructions, separate from the app's search boxes.

**Find the filter instructions.** The plan's condition row contains the `.sav` data source in column B, on row 4, 5, 6, or 7. Filters for columns appear on that same row. The statistic written in the table body is called `stat_type`: for example, `p` calculates a percentage, `n` a weighted count, and `n_unw` an unweighted count.

**Column B is global.** A filter such as `filter(age >= 18)` in B limits the whole table to adults. It remains active when another filter starts, when the statistic changes, and when a new table row begins. A later filter cannot bring excluded records back into the table.

**Additional filters apply to horizontal blocks.** In a horizontal table, a `filter(...)` in C or a later column combines with B. It applies to its own column and extends to the right. Its position relative to B makes no difference: it always combines with B. Changes from `p` to `n` or `n_unw` keep the filter, so percentages and their counts use the same population.

There are two ways an additional filter stops:

- **A non-mean statistic changes to a mean.** For example, when `n` is followed by `mean(age)`, that mean cell and following cells use B alone. Returning to `n` does not restart the old filter. This rule is checked across columns separately for each row; changes down rows do not stop it.
- **Another explicit filter starts.** The new filter replaces the previous additional filter and combines with B. It applies at its own starting column, including when that column calculates a mean.

Changes among non-mean statistics, between mean variables, and from a mean to a count do **not** stop the filter. A filter starting on a mean continues through later counts; a subsequent change from a non-mean to a mean stops it unless another explicit filter starts there.

Blank filter cells do not cancel an active filter. To remove an additional restriction explicitly, write `filter(TRUE)`; B still applies. To keep an additional restriction when entering a mean from a non-mean, repeat its `filter(...)` above that mean column.

**How to recognize the table direction.** The reader selects horizontal mode when the plan has `n_unw` immediately before `IDX`, or a `100/100.0` statistic. Other plans use vertical mode. In vertical tables, the first filter in B still restricts the common dataset; additional horizontal filter blocks are not processed. Check with the plan author before adding such blocks to a vertical table.

**Filter example: follow one row from left to right**

Suppose B contains `filter(age >= 18)`. In one horizontal table row, the instructions are:

| Column | Statistic    | Additional filter starts here    | Records available to this cell             |
|--------|--------------|----------------------------------|--------------------------------------------|
| C      | p            | None                             | All adults                                 |
| D      | p            | <code>filter(sex == 2)</code>    | Adults with sex code 2                     |
| E      | n            | None                             | Adults with sex code 2                     |
| F      | n_unw        | None                             | Adults with sex code 2                     |
| G      | mean(age)    | None                             | All adults: entered a mean from a non-mean |
| H      | n            | None                             | All adults: the old filter stays off       |
| I      | mean(income) | <code>filter(region == 1)</code> | Adults in region 1                         |
| J      | mean(age)    | None                             | Adults in region 1                         |
| K      | n_unw        | None                             | Adults in region 1                         |

No cell includes people under 18. D's filter applies to D-F, including the percentage and its counts. It stops at G. I starts a new filter, which applies to I-K and does not include D's sex restriction. Other row and column conditions still determine what is measured within these populations.

**Which changes count?** The engine uses the effective statistic after the worksheet reader fills missing statistic cells and ignores surrounding spaces when identifying means. Supported mean forms are `mean(variable)`, `mean`, and the legacy `Mean` form. A blank statistic cell does not itself mean "stop the filter." Changes such as p to n, n to n\_unw, `mean(age)` to `mean(income)`, and `mean(age)` to n all preserve it. Only entering a mean from a non-mean stops an inherited additional filter.

**When filters run.** For each required population, the engine loads the selected source, applies B, applies any active additional filter, and then runs the block's calculations. It calculates cell values from that data. The filters therefore affect both the numerator and denominator of a percentage. A filter variable must already exist before those calculations.

Only filter inheritance stops on entering a mean from a non-mean. Existing data-source, weight, calculation, and row/column-condition instructions keep their roles. A calculation-only instruction does not end the filter's column range.

**Filters do not reach left.** Adding a filter in G does not change F's calculated mean. Display suppression still needs an unweighted count: if a new filter separates a cell from its count, the engine uses the next n\_unw to its right on the same row and data source when no count is already linked. Existing count links and the small-sample thresholds remain in effect.

**Review after upgrading.** The earlier rule stopped additional filters at every statistic change, including p to n; this could give percentages and their counts different populations. Recalculate affected tables and review denominators and suppression. B remains global, including in later blocks; only non-mean-to-mean transitions now stop inherited filters.

## Choose decimal places in the worksheet

This section is for people who prepare or review the tabulation plan. Add **d=** inside a statistic's parentheses to choose how many decimal places are shown. This is available for every supported statistic type. If the statistic already has an argument, add a comma before **d=**.

| What you want                                  | What to write in the statistic cell | Example display |
|------------------------------------------------|-------------------------------------|-----------------|
| Whole-number average expenses in table 6.2a    | <code>mean(HCS8, d=0)</code>        | 442,572         |
| Average expenses with two decimals             | <code>mean(HCS8, d=2)</code>        | 442,571.91      |
| Mean indicator with one decimal                | <code>mean(d=1)</code>              | 73.3            |
| Percentage with two decimals                   | <code>p(d=2)</code>                 | 73.30           |
| Count with one decimal                         | <code>n(d=1)</code>                 | 881.0           |
| Unweighted count with no decimals              | <code>n_unw(d=0)</code>             | 622             |
| Median with two decimals                       | <code>median(age, d=2)</code>       | 35.00           |
| Total/distribution statistic with two decimals | <code>p(100, d=2)</code>            | 100.00          |

These numbers illustrate the display, not survey results. The same rule works with other supported statistics, including `mean_unw`, legacy `Mean`, and `p_sum`. A constant can be written as `100(d=2)`. Keep the statistic appropriate to the table's direction and intended calculation.

**If you leave out d, nothing changes:** the app keeps its established formatting for that statistic and output mode. Means default to one decimal place. For example, `mean(HCS8)` still displays 442,571.9.

**Use a whole number of decimal places.** `d=0` removes the decimal point; `d=1`, `d=2`, and `d=3` show one, two, and three decimal places, including trailing zeros. Spaces are allowed, so `mean(HCS8, d=0)` is valid. Give `d` only once, as a non-negative whole number, without quotation marks or a calculation. For example, `d=-1`, `d=1.5`, and `d="2"` are invalid.

**Where it applies.** The Formatted and Suppressed Tabulator views use the requested decimals for visible numbers. The Unformatted view retains the full numeric estimate. Both Excel output workbooks use the requested number format, including workbooks written by Multi-sheet Tabulator and values supplied by preparation scripts. Thousands separators remain in the Formatted preview and Excel. Long Format Data records the setting in `DECIMALS` and display values; `OBS_VALUE` retains full precision.

**What stays the same.** Display rounding does not change the calculated number, weights, filters, denominators, or consistency checks. Excel keeps numeric estimates at full precision when you specify `d`. Parenthesized estimates follow the requested precision, such as (442,572) for `d=0`; (\*) and - remain suppression markers.

Changing only `d` never starts or stops a filter. A column using `n_unw(d=2)` still acts as an unweighted count for direction and suppression. Where the reader fills a blank statistic cell, it carries the decimal setting with that statistic. Save the plan and reload it in the app, then rerun the affected tables to apply your edits.

## Other editors and code systems

The RStudio route is sufficient for the whole workflow. If you already use another environment, the same package installation and launch commands apply. These are alternative ways to start R, not separate versions of the app.

1. **Create or open a survey folder** using your editor's folder/workspace control. An `.Rproj` file is RStudio's project shortcut; other editors do not need it just to run R. You can open the same folder in another editor.
2. **Start an R session in that folder.** In Positron, select an R interpreter and use its Console. In VS Code, configure the R extension and use **R: Create R terminal**. In Cursor, use compatible R tooling or run R in its terminal if R is available there. In Jupyter, choose an R kernel; a Python kernel cannot run these commands. R's own console also works.
3. **Check the folder in R** with `getwd()` before installing or launching. Opening an editor folder does not always change an existing R session's working directory. Start a fresh R session in your survey folder if needed.
4. **Install the package** using the [installation commands](#) in that R session. Then check `here::here()` after installation: it shows the project root used for the default Excel output. If it selects a parent folder, supply your intended output directory explicitly when launching.
5. **Launch in a browser** with the command below, then [gather your inputs](#) and choose the [app tasks](#) you need:

```
micsPlusTableR::run_app(launch.browser = TRUE)
```

To choose the results folder explicitly, replace the example path below with an existing location on your computer. On Windows, use forward slashes, for example `C:/Users/YourName/Documents/MySurvey/results`:

```
micsPlusTableR::run_app(
  launch.browser = TRUE,
  output_dir = "/full/path/to/MySurvey/results"
)
```

**Keep R running:** the session or notebook cell stays busy while the app is open. Use its interrupt/stop control after saving your results. A terminal running R can usually be interrupted with **Ctrl+C**.

**Remote or hosted environments:** RStudio Server, Posit Workbench, and remote editor or notebook sessions need server-side package installation, access to the inputs, and a browser connection arranged by the administrator (often a proxy or port forwarding). Their output paths are on the server. The local 127.0.0.1 browser instructions assume R runs on your own computer; ask your support person for the hosted application's address and file-retrieval steps.

## Terms and reference information

### Words used in this guide

- **Click:** press the left mouse button once. On a touchpad, tap once.
- **Tab:** a section name across the top of the application.
- **Browse:** open a window to find a file on your computer.
- **Folder:** a place that holds files. A preparation folder may contain smaller folders.
- **Project:** a folder for one piece of survey work, with a file that helps RStudio reopen it.
- **Console:** the panel where you paste R commands and see messages.
- **Package:** an add-on for R; `micsPlusTableR` contains this application.
- **Tabulation plan:** an Excel workbook that specifies the tables to calculate, including their groups, conditions, and statistics.
- **Preparation script:** R code that prepares survey data and variables for the tabulation plan.
- **Sheet:** one page of an Excel workbook, such as 1.1a.
- **Wave:** one round of the survey. Use the wave your team gave you.

**About the pictures.** The Data Preparation picture shows Mongolia 2025-26 Wave 2. The Download Files picture shows the new tab in the 0.3.0 development version, before loading the survey list. The remaining pictures are screenshots from version 0.2.7, using Mongolia Wave 1 as an example. Your file names and results will differ. Use the files for your own survey. Screenshots show example output folders; your project normally saves to **micsPlusTableR-output**. In the HTML guide, click a picture to enlarge it; press **Esc** to return.

**Looking for package functions?** Use the [package reference manual](#) and [package documentation](#). Those documents cover programming and technical details.